

Amr ElSayyad

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Al-Khamael City, 6th Of October City, Giza, 12588, Egypt • Open to Remote •

Open to Relocate • [LinkedIn](#) • [GitHub](#) • [Portfolio](#)

Software Engineer

A skilled software engineer with a diverse technical background, including proficiency in various languages and a strong foundation in backend development. He has experience in building software solutions, interactive applications, and integrating third-party services.

EXPERIENCE

Giza Systems • 12/2024 — Present

Software Engineer – Machine Learning

Contributed to a telecommunications analytics platform designed for network performance monitoring, KPI computation, and real-time data visualization for telecom operators.

- Developed and maintained Python-based microservices using [Django](#) framework.
- Implemented high-performance data ingestion pipelines consuming from [Apache Kafka](#) with automated retry mechanisms and error handling for failed data processing.
- Contributed to the CI/CD using Docker, Kubernetes, and [ArgoCD](#).
- Managed Docker Compose environments for local development and testing
- Improved unit-testing pipelines using [Azure DevOps](#).
- Implemented [HashiCorp Vault](#) integration for secure configuration management.
- Created role-based IAM user management service using [Keycloak](#).

Orange Innovation Egypt (OIE) • 03/2023 — 12/2024

Software Engineer

- Contributed to [Open5GS](#), a C-language Open-Source implementation for 5G Core and EPC, implementing cutting edge solutions such as VoWiFi and LBO.
- Contributed to [Magma](#), a C, C++, Python, and Go Open-Source mobile core network solution, improving the usability of the software platform.
- Contributed to [SONiC](#), an Open-Source Network Operating System (NOS) based on Linux that runs on switches from multiple vendors and ASICs, improving security features.
- Contributed to a web UI for a [Network Services Orchestrator \(NSO\)](#) project using [Vaadin, a Full-Stack Java Platform for Building Modern Web Apps](#).
- Designed and implemented an inventory management system for the company using FastAPI for the backend and Angular for the frontend.

Schneider Electric Systems Egypt (SESE) • 03/2020 – 11/2021

Industrial Automation Engineer

- Represented the cybersecurity department in Egypt and supervised the IEC-62443-2-4 certification for industrial automation security.
- Designed the system architecture for a \$5 million project.
- Designed and implemented industrial automation solutions using Foxboro DCS and Triconex SIS for various clients and sectors.
- Supervised the assembly, testing, delivery, and maintenance of projects.

EDUCATION

Diploma in Artificial Intelligence

Information Technology Institute (ITI) • Egypt • 11/2021 – 08/2022

- Applied Data Engineering concepts to real-world data.
- Utilized Spark & PySpark for Big Data and Machine Learning.
- Applied Linear Algebra, Statistics, and Numerical Optimization for DS & ML.
- Developed AI applications using C++, Java, and Python.

B.Sc. in Mechatronics Engineering

October 6 University (O6U) • Egypt • (GPA: 3.59) • 09/2014 – 08/2019

- Graduation Project: Thermosyphon Water Purification System.
- Key Courses: Mathematics, Physics, Introduction to Computers, Production Technology, Mechanics, Thermodynamics, Stress Analysis, Electro-Magnetic Field, Logic Design, Engineering Analysis Methods, Electronics Circuit, Computer Aided Design, Power Electronics, Micro-Controller, Mechatronics Systems Design, Robotics, Principles of Scientific Thinking, Modeling and Simulation of Mechanical Systems, Artificial Intelligence.

SKILLS

Angular, Apache Kafka, Apache Spark, Artificial Intelligence, Azure DevOps, Bash, C, C++, CSS, Data Analysis, Data Science, Data Visualization, Django, Docker, Git, Hadoop, Hashicorp Vault, HTML, Java, JavaScript, JWT, Linux, Machine Learning, MATLAB, Maven, MS SQL Server, NoSQL, Plotly, PostgreSQL, PowerShell, Python, Redis, Spring Boot, Spring Framework, SQL, Tailwind CSS, T-SQL, TypeScript, Web Scraping

PROJECTS

Banking Portal API • [Link](#)

- Developed and maintained a Spring Boot app using JPA, Spring Security, and JWT.
- Utilized JUnit and Spring's MockMvc framework to simulate HTTP requests and validate API responses, enhancing test coverage and reliability.
- Leveraged Lombok for reducing boilerplate code.

Banking Portal UI • [Link](#)

- Developed and maintained a Front-End app using Angular, enhancing UI responsiveness.
- Implemented secure user authentication and authorization mechanisms, including JWT token management and email triggers for account activities.
- Integrated third-party libraries such as chart.js and ng2-charts for dynamic data visualization and ngx-countries-dropdown for improved country selection.
- Utilized Tailwind CSS for efficient and responsive UI design, ensuring cross-browser compatibility and mobile responsiveness.

Audio Stream • [Link](#)

- Developed a real-time audio streaming application in C++.
- Implemented functionality for recording, transmitting, receiving, and playing audio.
- Demonstrated proficiency in low-level audio programming and network communication.